1.1 SYSTEMS OF LINEAR EQUATIONS

Definition: A **linear equation** in the variables $x_1,\ldots,x_n$ is an equation that can be written in the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$ where b and the **coefficients** $a_1,\ldots,a_n$ are real or complex numbers.

- n may be any positive integer

- Some examples of linear equations are $6x_1 + 4x_2 - 3 = 7x_1$ $\qquad$ $x_2 = 3\left(\sqrt{5} - 2x_1\right) + x_3$

Definition: A **system of linear equations** (or a **linear system**) is a collection of one or more linear equations involving the same variables, say, $x_1,\ldots x_n$

- An example of a linear system is
$$\begin{aligned} 9x_1 - 3x_2 + 7.5x_3 &= 11 \\ x_1 \qquad\quad - 2x_3 &= -5 \end{aligned}$$

Definition: A **solution** of the system is a list $(s_1, s_2, \ldots, s_n)$ of numbers that makes each equation a true statement when the values of $s_1,\ldots,s_n$ are substituted for $x_1,\ldots x_n$, respectively.

Definition: The set of all possible solutions is called the **solution set** of the linear system.

Definition: Two linear systems are called **equivalent** if they have the same solution set.

**A system of linear equations has either

1. No Solution
2. Exactly One Solution
3. Infinitely Many Solutions

Ex 1: Determine h and k such that the solution set of the system (i) is empty, (ii) contains a unique solution, and (iii) contains infinitely many solutions. (Hold till end of 1.2 notes.)

$$\begin{aligned} x_1 + 3x_2 &= k \\ 4x_1 + hx_2 &= 8 \end{aligned}$$

Definition: A system of linear equations is said to be **consistent** if it has either one solution or infinitely many solutions; a system is **inconsistent** if it has no solutions.

MATRIX NOTATION

- Rectangular array called a **matrix**.
- Let us look at the following system and how we would write two matrices called the coefficient matrix and the augmented matrix for it.

$$\begin{aligned} x_1 - 2x_2 + 5x_3 &= 6 \\ 3x_1 \qquad - 4x_3 &= 0 \\ - 6x_2 - x_3 &= 4 \end{aligned}$$

Definition: A **<u>coefficient matrix</u>** consists of the coefficients of each variable aligned in columns. $\begin{bmatrix} 1 & -2 & 5 \\ 3 & 0 & -4 \\ 0 & -6 & -1 \end{bmatrix}$

Definition: An **<u>augmented matrix</u>** consists of the coefficient matrix with an added column containing the constants from the right sides of the equation. $\begin{bmatrix} 1 & -2 & 5 & 6 \\ 3 & 0 & -4 & 0 \\ 0 & -6 & -1 & 4 \end{bmatrix}$

Definition: The **<u>size</u>** of a matrix tells how many rows and columns are in the matrix, such as a 2 X 5 (read 2 by 5) has 2 rows and 5 columns.
- If m and n are positive integers, an **<u>m x n matrix</u>** is a rectangular array of numbers with **<u>m rows</u>** and **<u>n columns</u>**.

<u>SOLVING A LINEAR SYSTEM</u>
- the basic strategy is to replace one system with an equivalent system (i.e., one with the same solution set) that is easier to solve.

3 basic operations are used to simplify a linear system
- Replace one equation by the sum of itself and a multiple of another equation.
- Interchange two equations.
- Multiply all the terms in an equation by a nonzero constant.

<u>Ex 2</u>: Solve the following system

$$\begin{aligned} x_1 - 3x_2 &= 5 \\ -x_1 + x_2 + 5x_3 &= 2 \\ x_2 + x_3 &= 0 \end{aligned}$$

- 3 basic operations listed earlier correspond to the following operations on the augmented matrix.

<u>Elementary Row Operations</u>
1. (**<u>Replacement</u>**) Replace one row by the sum of itself and a multiple of another row.
2. (**<u>Interchange</u>**) Interchange two rows.
3. (**<u>Scaling</u>**) Multiply all entries in a row by a nonzero constant

- Row operations can be applied to any matrix.

Definition: Two matrices are **<u>row equivalent</u>** if there is a sequence of elementary row operations that transforms one matrix into the other.
- If the augmented matrices of two linear systems are row equivalent, then, the two systems have the same solution set.

<u>EXISTENCE & UNIQUENESS QUESTIONS</u>

Two Fundamental Questions About a Linear System
1. Is the system consistent, that is, does at least one solution exist?
2. If a solution exists, is it the only one; that is, is the solution unique?

- Like to find the answer without completely solving the linear system.

Ex 3: Is the system consistent? Do not completely solve the system.

$$\begin{aligned} x_1 \qquad\qquad & -2x_4 = -3 \\ 2x_2 + 2x_3 \qquad & = 0 \\ x_3 + 3x_4 & = 1 \\ -2x_1 + 3x_2 + 2x_3 + x_4 & = 5 \end{aligned}$$

- An example of how a row of an inconsistent matrix would look like is this: 0 0 0 -4 which is the same as $0x_1 + 0x_2 + 0x_3 = -4$.

1.2 ROW REDUCTION AND ECHELON FORMS

Definition: A **nonzero** row or column in a matrix, means a row or column that contains at least one nonzero entry.
Definition: A **leading entry** of a row refers to the leftmost nonzero entry (in a nonzero row).
Definition: A rectangular matrix is in **echelon form** (or **row echelon** form) if it has the following three properties:

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

Definition: If a matrix in echelon form satisfies the following additional conditions, then it is in **reduced echelon form** (or **reduced row echelon** form):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column

Theorem 1: **Uniqueness of the Reduced Echelon Form** – Each matrix is row equivalent to one and only one reduced echelon form.

Calculator - REF – row echelon form RREF – reduced row echelon form

Definition: A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A.

Definition: A **pivot column** is a column of A that contains a pivot position.

- Earlier, the squares ( ) identify the pivot positions.

Ex 4: Circle the pivot positions in the final matrix and in the original matrix, and list the pivot columns.

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix}$$

Definition: A **pivot** is a nonzero number in a pivot position that is used as needed to create zeros via row operations.

ROW REDUCTION ALGORITHM

Step 1: Begin with the left most nonzero column. This is a pivot column. The pivot position is at the top.
Step 2: Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position.
Step 3: Use row replacement operations to create zeros in all positions below the pivot.
Step 4: Cover (or ignore) the row containing the pivot position and cover all rows, if any above it. Apply steps 1 – 3 to the submatrix that remains. Repeat the process until there are no more nonzero rows to modify.
Step 5: Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot. If a pivot is not 1, make it 1 by a scaling operation.

- Steps 1 – 4 are called the **forward phase**.
- Step 5 is called the **backward phase**.
- The result of the first 4 steps leaves the matrix in echelon form. The fifth step leaves the matrix in reduced echelon form.

Ex 5: Use the Row Reduction Algorithm to solve the system of equations.

$$\begin{aligned} x_1 + x_2 + 2x_3 &= -1 \\ -x_1 \quad\quad + 2x_3 &= -3 \\ \quad x_2 + 2x_3 &= -2 \\ 2x_1 - x_2 + x_3 &= 1 \end{aligned}$$

SOLUTIONS OF LINEAR SYSTEMS

Just an example: $\begin{bmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ = $\begin{aligned} x_1 \quad & -3x_3 = -1 \\ x_2 - & x_3 = 6 \\ 0 & = 0 \end{aligned}$

x_1 and x_2 are called **basic variables**. x_3 is called a **free variable**.

General Solution is $\begin{cases} x_1 = -1 + 3x_3 \\ x_2 = 6 + x_3 \\ x_3 \text{ is free} \end{cases}$

Parametric Description of Solution Sets

- The description in the previous part (General Solution) are parametric descriptions of solution sets in which the free variables act as parameters.
- Solving a system amounts to finding a parametric description of the solution set or determining that the solution set is empty.

EXISTENCE AND UNITQUENESS

Theorem – A linear system is consistent if and only if the rightmost column – that is, if and only if an echelon form of the augmented matrix has no row of the form [0 . . . 0 b] with b nonzero.

If a linear system is consistent then the solution set contains wither (1) a unique solution, where there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.
In Ex 6 find the general solutions of the systems.

Ex 6: $\begin{bmatrix} 1 & 4 & 0 & 7 \\ 2 & 7 & 0 & 10 \end{bmatrix}$

Ex 7: $\begin{bmatrix} 1 & -7 & 0 & 6 & 5 \\ 0 & 0 & 1 & -2 & -3 \\ -1 & 7 & -4 & 2 & 7 \end{bmatrix}$

USING ROW REDUCTION TO SOLVE A LINEAR SYSTEM

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If there is no solution, stop; otherwise, go to the next step.
3. Continue row reduction to obtain the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in step 3.
5. Rewrite each nonzero equation from step 4 so that its one basic variable is expressed in terms of any free variables appearing in the equation.

1.3 VECTOR EQUATIONS

Vectors in $\mathbb{R}^2$

Column Vector $u = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ $v = \begin{bmatrix} -7 \\ 0 \end{bmatrix}$ $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ where w_1 and w_2 are any real numbers

- Two vectors are equal if and only if their corresponding entries are equal.

Sum	**u + v**	add corresponding entries
Scalar Multiple	**cu**	multiply each entry by the same constant, where c is a **scalar**.

Ex 1: $u = \begin{bmatrix} 2 \\ -7 \end{bmatrix}$ $v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ Find (a) (-3)**v** (b) 2**u** (c) 2**u** + (-3)**v**

Geometric Descriptions of $\mathbb{R}^2$

- geometric point (a, b) as a column vector $\begin{bmatrix} a \\ b \end{bmatrix}$

Parallelogram Rule for Addition

If **u** and **v** in $\mathbb{R}^2$ are represented as points in the plane, then **u + v** corresponds to the fourth vertex of the parallelogram whose other vertices are **0**, **u**, and **v**.

Vectors in $\mathbb{R}^3$

- 3 x 1 column matrices with three entries
- Geometrically: points in a 3-D coordinate space, with arrows from the origin.

Vectors in $\mathbb{R}^n$

- n is a positive integer, $\mathbb{R}^n$ lists of n real numbers $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ - Vectors with all zeros is called the zero vector

ALGEBRAIC PROPERTIES OF $\mathbb{R}^n$

- For all **u**, **v**, **w** in $\mathbb{R}^n$ and all scalars c and d:

(i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
(ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
(iii) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
(iv) $\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$
where $-\mathbf{u}$ denotes $(-1)\mathbf{u}$
(v) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
(vi) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
(vii) $c(d\mathbf{u}) = (cd)\mathbf{u}$
(viii) $1\mathbf{u} = \mathbf{u}$

LINEAR COMBINATIONS

Given vectors $v_1, v_2, \ldots, v_p$ in $\mathbb{R}^n$ and given scalars $c_1, c_2, \ldots, c_p$, vector **y** defined by $y = c_1v_1 + c_2v_2 + \cdots + c_pv_p$ is called a **linear combination** of $v_1, v_2, \ldots, v_p$ with **weights** $c_1, c_2, \ldots, c_p$.

- Page 28 – Shows an illustration of linear combinations of two vectors using parallel grid lines

Ex 2: Write a system of equations that is equivalent to: $x_1\begin{bmatrix}-2\\3\end{bmatrix} + x_2\begin{bmatrix}8\\5\end{bmatrix} + x_3\begin{bmatrix}1\\-6\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$

Ex 3: Write a vector equation that is equivalent to:

$$\begin{aligned} 4x_1 + x_2 + 3x_3 &= 9 \\ x_1 - 7x_2 - 2x_3 &= 2 \\ 8x_1 + 6x_2 + 5x_3 &= 15 \end{aligned}$$

**A vector equation $x_1a_1 + x_2a_2 + \cdots + x_na_n = b$ x has the same solution set as the linear system whose augmented matrix is $\begin{bmatrix} a_1 & a_2 & \cdots & a_n & b \end{bmatrix}$.

- In particular, **b** can be generated by a linear combination of $a_1, \ldots, a_n$ if and only if there exists a solution to the linear system corresponding to $\begin{bmatrix} a_1 & a_2 & \cdots & a_n & b \end{bmatrix}$.

Ex 4: Determine if **b** is a linear combination of a_1, a_2, and a_3.

$$a_1 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \quad a_2 = \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix} \quad a_3 = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix} \quad b = \begin{bmatrix} -5 \\ 11 \\ -7 \end{bmatrix}$$

Ex 5: Determine if **b** is a linear combination of the vectors formed from the columns of the matrix A.

$$A = \begin{bmatrix} 1 & -2 & -6 \\ 0 & 3 & 7 \\ 1 & -2 & 5 \end{bmatrix}, \quad b = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$$

Definition: If $v_1, v_2, \ldots, v_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $v_1, v_2, \ldots, v_p$ is denoted by $Span\{v_1, \ldots, v_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned** (or **generated**) **by** $v_1, v_2, \ldots, v_p$. That is, $Span\{v_1, \ldots, v_p\}$ is the collection of all vectors that can be written in the form $c_1v_1 + c_2v_2 + \cdots + c_pv_p$ with $c_1, \ldots, c_p$ scalars.

- Asking whether a vector b is in $Span\{v_1, \ldots, v_p\}$ amounts to asking whether the vector equation $x_1v_1 + x_2v_2 + \cdots + x_pv_p = b$ has a solution, or, equivalently, asking whether the linear system with augmented matrix $\begin{bmatrix} v_1 & v_2 & \cdots & v_p & b \end{bmatrix}$ has a solution.

- Note that $Span\{v_1, \ldots, v_p\}$ contains every scalar multiple of v_1 (for example), since $cv_1 = cv_1 + 0v_2 + \cdots + 0v_p$. In particular, the zero vector must be in $Span\{v_1, \ldots, v_p\}$.

A Geometric Description of Span {v} and Span {u, v}

- Let **v** be a nonzero vector in $\mathbb{R}^3$. Then Span {**v**} is the set of all scalar multiples of **v**, and we visualize it as the set of points on the line in $\mathbb{R}^3$ through **v** and **0**.

- **u** and **v** nonzero vectors in $\mathbb{R}^3$, with **v** not a multiple of **u**, Span {**u**, **v**} is a plane in $\mathbb{R}^3$ that contains **u**, **v** and **0**. Line in $\mathbb{R}^3$ through **u** and **0** and **v** and **0**. (Page 35)

Ex 6: Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, $v_2 = \begin{bmatrix} -3 \\ 1 \\ 8 \end{bmatrix}$, and $y = \begin{bmatrix} h \\ -5 \\ -3 \end{bmatrix}$. For what value(s) of h is y in the plane generated by v_1 and v_2?

Ex 7: Let $A = \begin{bmatrix} 2 & 0 & 6 \\ -1 & 8 & 5 \\ 1 & -2 & 1 \end{bmatrix}$, let $b = \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix}$, and let W be the set of all linear combinations of the columns of A.

(a) Is b in W?

(b) Show that the third column of A is in W.

1.4 The Matrix Equation Ax = b

- A fundamental idea in linear algebra is to view a linear combination of vectors as the product of a matrix and a vector.

Definition: If A is an m x n matrix, with columns $a_1,\ldots,a_n$, and if x is in $\mathbb{R}^n$, then the **product of A and x**, denoted by A**x**, is **the linear combination of the columns of A using the corresponding entries in x as weights**; that is,

$$A\mathbf{x} = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1a_1 + x_2a_2 + \cdots + x_na_n$$

- A**x** is defined only if the number of columns of A equals the number of entries in **x**.

Ex 1: $\begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix} = ?$

Ex 2: Write the linear combination $-2v_1 + 4v_2 - 8v_3$ as a matrix times a vector.

- Earlier, we learned $\begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \end{aligned}$ as $x_1\begin{bmatrix} 1 \\ -2 \end{bmatrix} + x_2\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$. Now, write it as $\underset{A}{\begin{bmatrix} 1 & 2 \\ -2 & 5 \end{bmatrix}} \underset{x}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} = \underset{b}{\begin{bmatrix} 7 \\ 4 \end{bmatrix}}$.

Theorem 3 – If A is an m x n matrix, with columns $a_1,\ldots,a_n$, and if b is in $\mathbb{R}^n$, the matrix equation A**x** = **b** has the same solution set as the vector equation $x_1a_1 + x_2a_2 + \cdots + x_na_n = b$ which, in turn, has the same solution set as the system of linear equations whose augmented matrix $\begin{bmatrix} a_1 & a_2 & \cdots & a_n & b \end{bmatrix}$.

- We can view a system of linear equations in three different but equivalent ways:
  matrix equation vector equation system of linear equations

- All solved in the same way – by row reducing the augmented matrix.

Existence of Solutions

The equation A**x** = **b** has a solution if and only if **b** is a linear combination of the columns of A.
- In 1.3, we looked at the question, "Is **b** in $\text{Span}\{a_1,\ldots,a_n\}$?" which is equivalent to "Is A**x** = **b** consistent?"

- A harder existence problem is to determine whether the equation A**x** = **b** is consistent for all possible **b**.

Ex 3: Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -3 & 2 & 6 \\ 5 & -1 & -8 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Show that the equation A**x** = **b** **does not** have a solution for all possible b, and describe the set of all **b** for which A**x** = **b** **does** have a solution.

- In the next theorem, **the columns of A span** $\mathbb{R}^m$, mean that every **b** in $\mathbb{R}^m$ is a linear combination of the columns of A.
- In general, a set of vectors $\{v_1, \ldots, v_p\}$ in $\mathbb{R}^m$ **spans** (or **generates**) $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $v_1, \ldots, v_p$, that is, if $\text{Span}\{v_1, \ldots, v_p\} = \mathbb{R}^m$.

Theorem 4 – Let A be an m x n matrix. Then the following statements are logically equivalent. That is, for a particular A, either they are all true statements or they are all false.

a. For each b in $\mathbb{R}^m$, the equation A**x** = **b** has a solution.
b. Each **b** in $\mathbb{R}^m$ is a linear combination of the columns of A.
c. The columns of A span $\mathbb{R}^m$.
d. A has a pivot position in every row.

WARNING: **Theorem 4** is about a **coefficient matrix**, not an augmented matrix. If an augmented matrix [A b] has a pivot position in every row, then the equation A**x** = **b** may or may not be consistent.

Ex 4: Let $v_1 = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ -3 \\ 8 \end{bmatrix}$, $v_3 = \begin{bmatrix} 4 \\ -1 \\ -5 \end{bmatrix}$. Does $\{v_1, v_2, v_3\}$ span $\mathbb{R}^3$? Why or why not?

Computation of Ax

If the product A**x** is defined, then the ith entry in A**x** is the sum of the products of corresponding entries from row i of A and from the vector **x**.

- **Identity Matrix** – **I** – 1's on the diagonal and 0's elsewhere. $I_n x = x$ for every x in $\mathbb{R}^n$

Properties of the Matrix-Vector Product Ax

Theorem 5 – If A is an m x n matrix, **u** and **v** are vectors in $\mathbb{R}^n$, and c is a scalar, then:

(a) A (**u** + **v**) = A**u** + A**v**

(b) A (c**u**) = c (A**u**)

Ex 5: Determine if the columns of the matrix span $\mathbb{R}^4$. $\begin{bmatrix} 5 & -7 & -4 & 9 \\ 6 & -8 & -7 & 5 \\ 4 & -4 & -9 & -9 \\ -9 & 11 & 16 & 7 \end{bmatrix}$

Ex 6: Determine if the columns of the matrix span $\mathbb{R}^4$. $\begin{bmatrix} 8 & 11 & -6 & -7 & 13 \\ -7 & -8 & 5 & 6 & -9 \\ 11 & 7 & -7 & -9 & -6 \\ -3 & 4 & 1 & 8 & 7 \end{bmatrix}$.

Ex 7: Write the augmented matrix for the linear system that corresponds to the matrix equation A**x** = **b**. Then solve the system and write the solution as a vector. $A = \begin{bmatrix} 1 & 2 & 1 \\ -3 & -1 & 2 \\ 0 & 5 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$

Discussion Questions

1. Let A be a 3 x 2 matrix. Explain why the equation A**x** = **b** cannot be consistent for all **b** in $\mathbb{R}^3$. Generalize your argument to the case of an arbitrary A with more rows than columns.

2. Could a set of three vectors in $\mathbb{R}^4$ span all of $\mathbb{R}^4$? Explain. What about *n* vectors in $\mathbb{R}^m$ when *n* is less than *m*?

1.5 SOLUTION SETS OF LINEAR SYSTEMS

Homogeneous Linear Systems A system of linear equations is said to be homogeneous if it can be written in the form $A\mathbf{x} = \mathbf{0}$, where A is an *m* x *n* matrix and **0** is the zero vector in R^m.

For example, $x_1 + 10x_2 = 0$
$2x_1 + 20x_2 = 0$ is a homogeneous system.

Its corresponding matrix equation $A\mathbf{x} = \mathbf{0}$ is $\begin{bmatrix} 1 & 10 \\ 2 & 20 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

The homogeneous system $A\mathbf{x} = \mathbf{0}$ always has at least one solution, namely $\mathbf{x} = \mathbf{0}$.

The zero solution is usually called the trivial solution.

The important question is whether there exists a nontrivial solution, that is, a nonzero vector **x** that satisfies $A\mathbf{x} = \mathbf{0}$.

The homogeneous equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution if and only if the equation has at least one free variable.

Do nontrivial solutions exist for our system $\begin{bmatrix} 1 & 10 \\ 2 & 20 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$? Note $\begin{bmatrix} 1 & 10 \\ 2 & 20 \end{bmatrix} \sim \begin{bmatrix} 1 & 10 \\ 0 & 0 \end{bmatrix}$

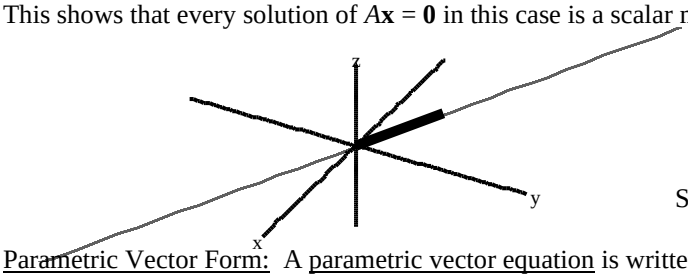
So, it is a consistent system with a free variable and has infinitely many solutions.

Ex 1: Does the system have a nontrivial solution? Try to use as few row operations as possible.

$$-3x_1 + 5x_2 - 7x_3 = 0$$
$$-6x_1 + 7x_2 + x_3 = 0$$

Graphical representation

This shows that every solution of $A\mathbf{x} = \mathbf{0}$ in this case is a scalar multiple of **v.** The trivial solution is achieved by choosing $\boldsymbol{x}_2 = \mathbf{0}$.



Solution set = span{**v**} = line through **0** in R^3

Parametric Vector Form: A parametric vector equation is written as $\mathbf{x} = s\mathbf{u} + t\mathbf{v}$ (s, t in R)
Parametric vector equation of a line $\mathbf{x} = t\mathbf{v}$ (with t in R)

The following are in parametric vector form. $x_3\begin{bmatrix}2\\0\\1\end{bmatrix}$ or $x_2\begin{bmatrix}-3\\6\\1\end{bmatrix} + x_3\begin{bmatrix}1\\0\\4\end{bmatrix}$

Solutions of Nonhomeogeneous Systems

When a nonhomogeneous linear system has many solutions, the general solution can be written in parametric vector form as one vector plus an arbitrary linear combination of vectors that satisfy the corresponding homogeneous system.

Ex 2: Write the solution set in parametric vector form of

$$x_1 + 3x_2 - 5x_3 = 0$$
$$x_1 + 4x_2 - 8x_3 = 0$$
$$-3x_1 - 7x_2 + 9x_3 = 0$$

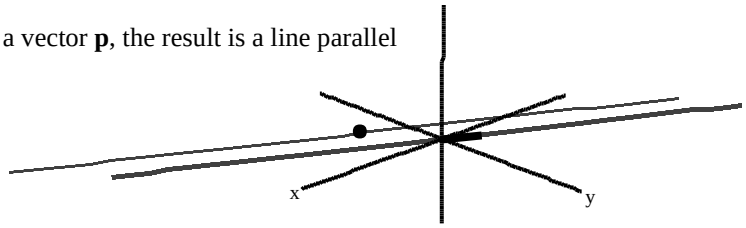
Ex 3: Describe all solutions of $A\mathbf{x} = \mathbf{0}$ in parametric vector form, where A is row equivalent to $\begin{bmatrix}1 & -2 & -9 & 5\\0 & 1 & 2 & -6\end{bmatrix}$

To describe the solution set of $A\mathbf{x} = \mathbf{b}$ geometrically, think of vector addition as a translation.

Given $\mathbf{v}$ and $\mathbf{p}$ in R^2 or R^3, the effect of adding $\mathbf{p}$ to $\mathbf{v}$ is to move $\mathbf{v}$ in a direction parallel to the line through $\mathbf{p}$ and $\mathbf{0}$. We say $\mathbf{v}$ is translated by $\mathbf{p}$ to $\mathbf{v} + \mathbf{p}$.

If each point on a line L in R^2 or R^3 is translated by a vector $\mathbf{p}$, the result is a line parallel to L.

Note that the solution sets of $A\mathbf{x} = \mathbf{0}$ and $A\mathbf{x} = \mathbf{b}$ are parallel.



Suppose L is the line through $\mathbf{0}$ and $\mathbf{v}$,
$\mathbf{x} = t\mathbf{v}$ (t in$R$). Adding $\mathbf{p}$ to each point on L produces the line $\mathbf{x} = \mathbf{p} + t\mathbf{v}$. $\mathbf{x} = \mathbf{p} + t\mathbf{v}$ is the equation of the line through $\mathbf{p}$ parallel to $\mathbf{v}$.

Theorem 6 : Suppose the equation $A\mathbf{x} = \mathbf{b}$ is consistent for some given $\mathbf{b}$, and let $\mathbf{p}$ be a solution. Then the solution set of $A\mathbf{x} = \mathbf{b}$ is the set of all vectors of the form $\mathbf{w} = \mathbf{p} + \mathbf{v}_h$ where $\mathbf{v}_h$ is any solution of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Theorem 6 says that if $A\mathbf{x} = \mathbf{b}$ has a solution, then the solution set is obtained by translating the solution set of $A\mathbf{x} = \mathbf{0}$, using any particular solution p of $A\mathbf{x} = \mathbf{b}$ for the translation.

WARNING: Theorem 6 applies only to an equation $A\mathbf{x} = \mathbf{b}$ that has at least one nonzero solution $\mathbf{p}$. When $A\mathbf{x} = \mathbf{b}$ has no solution, the solution set is empty.

Ex 4: Find a parametric equation of the line through $\mathbf{a}$ parallel to $\mathbf{b}$. $\mathbf{a} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -7 \\ 8 \end{bmatrix}$

Recap:Writing A Solution Set (of a Consistent System) In Parametric Vector Form

1. Row reduce the augmented matrix to reduced echelon form.
2. Express each basic variable in terms of any free variables appearing in an equation.
3. Write a typical solution $\mathbf{x}$ as a vector whose entries depend on the free variables, if any.
4. Decompose $\mathbf{x}$ into a linear combination of vectors (with numeric entries) using the free variables as parameters.

Ex 5: Solve the system of equations.

$$\begin{aligned} x_1 + 2x_2 \quad & -3x_4 + \quad x_5 \quad = 2 \\ x_1 + 2x_2 + x_3 & -3x_4 + \quad x_5 + 2x_6 = 3 \\ x_1 + 2x_2 \quad & -3x_4 + 2x_5 + \quad x_6 = 4 \\ 3x_1 + 6x_2 + x_3 & -9x_4 + 4x_5 + 3x_6 = 9 \end{aligned}$$

1.6 Applications of Linear Systems

Linear systems with many solutions can arise naturally, such as in economics, chemistry, and network flow.

Economics - A nation's economy is divided into many sectors, such as manufacturing, communication, entertainment and service industries.

Suppose for each sector we know the total output for one year and exactly how this output is divided or "exchanged" among the other sectors of the economy. Let the total dollar value of a sector's output be called the price of that output.

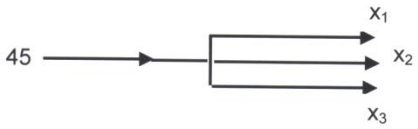
There exist equilibrium prices that can be assigned to the total outputs of the various sectors in such a way that the income of each sector exactly balances its expenses.

In order to find equilibrium prices, set up a matrix where each sector looks down a column to see where its output goes and look across a row to see what it needs as inputs. The decimal fractions in each column sum to 1. (You will get a practice WS on this too.)

Network Flow is the study of the flow of some quantity through a network such as monitoring the pattern of traffic flow in a grid of city streets or the current flow through electrical circuits.

A network consists of a set of points called junctions, or nodes, with lines or arcs called branches connecting some or all of the junctions.
The direction of flow in each branch is indicated, and the flow amount (or rate) is either shown or is denoted by a variable.

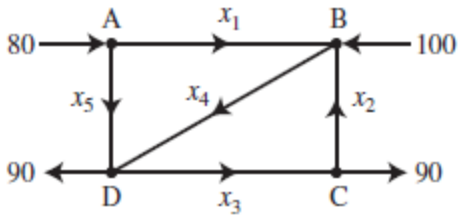
The key assumption of network flow is that the total flow into the network equals the total flow out of the network and that the total flow into a junction equals the total flow out of the junction.



$45 = x_1 + x_2 + x_3$

Ex 1: (#14 page 55)

(a) Find the general traffic pattern in the freeway network (Rate in cars per minute.)

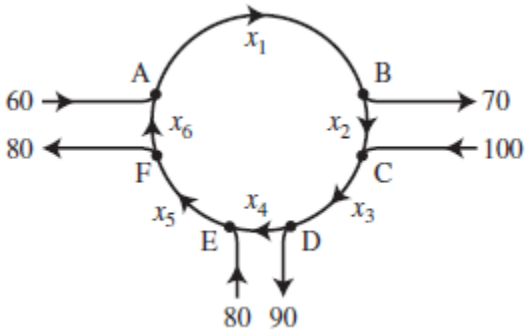


(b) Describe the general traffic pattern when the road whose flow is x_5 is closed.

(c) When $x_5 = 0$, what is the minimum value of x_4?

Ex 2: (#15 Page 55)
Intersections in England are often constructed as one-way "roundabouts", such as the one shown. Assume that traffic must travel in the directions shown.

(a) Find the general solution of the network flow.



(b) Find the smallest possible value for x_6.

1.7 LINEAR INDEPENDENCE

The homogeneous equations can be studied from a different perspective by writing them as vector equations.

For example, $\begin{bmatrix} 1 & 2 & -3 \\ 3 & 5 & 9 \\ 5 & 9 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ can be written as the vector equation $x_1\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + x_2\begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix} + x_3\begin{bmatrix} -3 \\ 9 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Definition: An indexed set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in R^n is said to be linearly independent if the vector equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots x_p\mathbf{v}_p = \mathbf{0}$ has only the trivial solution. The set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is said to be linearly dependent if there exist weights $c_1, \ldots, c_p$, not all zero, such that $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots c_p\mathbf{v}_p = \mathbf{0}$.

The last equation is called a linear dependence relation among $\mathbf{v}_1, \ldots, \mathbf{v}_p$ when the weights are not all zero.

An indexed set is linearly dependent if and only if it is not linearly independent.

In Ex 1 and 2, determine if the vectors are linearly independent. Justify each answer.

Ex 1: $\begin{bmatrix}0\\0\\2\end{bmatrix}, \begin{bmatrix}0\\5\\-8\end{bmatrix}, \begin{bmatrix}-3\\4\\1\end{bmatrix}$

Ex 2: $\begin{bmatrix}1\\-3\end{bmatrix}, \begin{bmatrix}-3\\9\end{bmatrix}$

Linear Independence of Matrix Columns

Begin with a matrix $A = [a_1 \ldots a_n]$ instead of set of vectors. $A\mathbf{x} = \mathbf{0}$ implies . $x_1a_1 + x_2a_2 + \cdots + x_na_n = 0$

The columns of matrix A are linearly independent if and only if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
In Ex 3 and 4, determine if the columns of the matrix form a linearly independent set. Justify your answer.

Ex 3: $\begin{bmatrix}0 & -8 & 5\\3 & -7 & 4\\-1 & 5 & -4\\1 & -3 & 2\end{bmatrix}$

Ex 4: $\begin{bmatrix}1 & -3 & 3 & -2\\-3 & 7 & -1 & 2\\0 & 1 & -4 & 3\end{bmatrix}$

Ex 5: $v_1 = \begin{bmatrix}1\\-5\\-3\end{bmatrix}, v_2 = \begin{bmatrix}-2\\10\\6\end{bmatrix}, v_3 = \begin{bmatrix}2\\-9\\h\end{bmatrix}$

(a) For what values of h is $\mathbf{v}_3$ in Span$\{\mathbf{v}_1, \mathbf{v}_2\}$

(b) For what values of h is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly dependent?

Special Cases Sometimes we can determine linear independence of a set with minimal effort.

1. Set of One Vector

A set containing only vector, $\mathbf{v}$, is linearly independent if and only if $\mathbf{v}$ is not the zero vector.

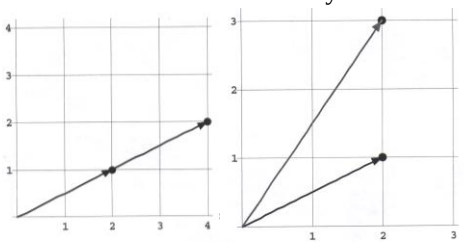
Consider the set containing one nonzero vector $\{\mathbf{v}_1\}$. The only solution to $x_1\,\mathbf{v}_1 = 0$ is $x_1 = ?$

So, $\{\mathbf{v}_1\}$. Is linearly independent when $\mathbf{v}_1 \neq 0$

2. A Set of Two Vectors

A set of two vectors $\{\mathbf{v}_1,\mathbf{v}_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other. The set is linearly independent if and only if neither of the vectors is a multiple of the other.

Let $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$



One of these pairs is linearly dependent and one is linearly independent. Which is which?

In geometric terms, two vectors are linearly dependent if and only if they lie on the same line through the origin.

Sets of Two or More Vectors

Theorem 7: Characterization of Linearly Dependent Sets: An indexed set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others.

In fact, if S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors, $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

WARNING – Theorem 7 does not say that every vector in a linearly dependent set is a linear combination of the preceding vectors. A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

Theorem 8**:** If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in R^n is linearly dependent if $p > n$.

WARNING – Theorem 8 says nothing about the case when the number of vectors in the set does NOT exceed the number of entries in each vector.

Theorem 9: If a set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in R^n contains the zero vector, then the set is linearly dependent.

In Ex 6 – 9, determine by inspection whether the vectors are linearly independent. Justify your answer.

Ex 6: $\begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}, \begin{bmatrix} 6 \\ -3 \\ 9 \end{bmatrix}$

Ex 7: $\begin{bmatrix} 4 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 8 \\ 1 \end{bmatrix}$

Ex 8: $\begin{bmatrix} -8 \\ 12 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}$

Ex 9: $\begin{bmatrix} 1 \\ 4 \\ -7 \end{bmatrix}, \begin{bmatrix} -2 \\ 5 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

1.8 Introduction to Linear Transformations

Another way to view $A\mathbf{x} = \mathbf{b}$ is to view matrix A as an object acting on $\mathbf{x}$ by multiplication to produce a new vector $A\mathbf{x}$ or $\mathbf{b}$.

$\mathbf{x}$

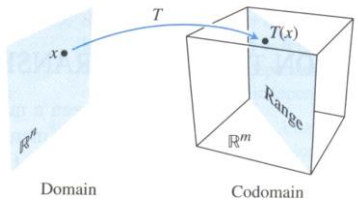
Transformation $T : \mathbf{R}^n \rightarrow \mathbf{R}^m$ *machine*

$T(\mathbf{x})$

For example, if $A = \begin{bmatrix} 2 & -4 \\ 3 & -6 \\ 1 & -2 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, then $A\mathbf{u}$ transforms the 2 x 1 vector to the 3 x 1 vector $\begin{bmatrix} -8 \\ -12 \\ -4 \end{bmatrix}$.

How does A transform $\mathbf{v}$?

A transformation (or function or **mapping**) T from R^n to R^m is a rule that assigns to each vector $\mathbf{x}$ in R^n a vector $T(\mathbf{x})$ in R^m. The set R^n is called the domain of T and R^m is called the codomain of T. $T: R^n \rightarrow R^m$ .For $\mathbf{x}$ in R^n, the vector $T(\mathbf{x})$ in R^m is called the image of $\mathbf{x}$ (under the action of T). The set of all images $T(\mathbf{x})$ is called the range of T.



Matrix Transformations

For each $\mathbf{x}$ in R^n, $T(\mathbf{x})$ is computed as $A\mathbf{x}$ where A is a m x n matrix.

Matrix transformation $\mathbf{x} \rightarrow A\mathbf{x}$

Domain of T is R^n when A has n columns and the codomain of T is R^m when each column of A has m entries.

The range of T is the set of all linear combinations of the columns of A, because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$.

In Ex 1 and Ex 2, find a vector $\mathbf{x}$, whose image under T is $\mathbf{b}$, and determine whether $\mathbf{x}$ is unique.

Ex 1: $A = \begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & -4 \\ 3 & -5 & -9 \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ -7 \\ -9 \end{bmatrix}$

Ex 2: $A = \begin{bmatrix} 1 & -2 & 1 \\ 3 & -4 & 5 \\ 0 & 1 & 1 \\ -3 & 5 & -4 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 9 \\ 3 \\ -6 \end{bmatrix}$

Ex 3: Let $b = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 4 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & 3 & 9 & 2 \\ 1 & 0 & 3 & -4 \\ 0 & 1 & 2 & 3 \\ -2 & 3 & 0 & 5 \end{bmatrix}$

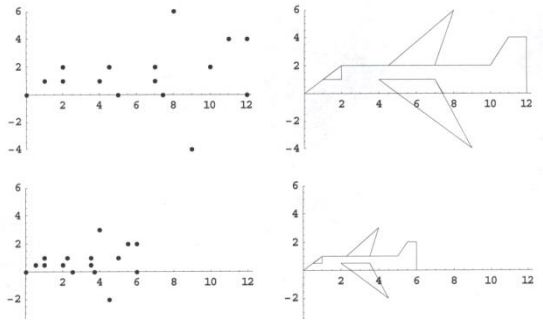
Is $\mathbf{b}$ in the range of the linear transformation $\mathbf{x} \rightarrow A\mathbf{x}$? Why or why not?

Matrix transformations have many applications - including computer graphics.

Ex 4: Let $A=\begin{bmatrix}.5 & 0 & 0\\ 0 & .5 & 0\\ 0 & 0 & .5\end{bmatrix}$, $u=\begin{bmatrix}1\\ 0\\ -4\end{bmatrix}$, $v=\begin{bmatrix}a\\ b\\ c\end{bmatrix}$.

Define $T: R^3 \to R^3$ by $T(\mathbf{x}) = A\mathbf{x}$.

Find T(**u**) and T(**v**).



The previous transformation is an example of a contraction transformation. It can be used to move a point or set of points. In a plane, you can transform a diagram as shown.

$T: R^2 \to R^2$ by $T(\mathbf{x}) = r\mathbf{x}$ T is a contraction when $0 \le r \le 1$ and a dilation when r > 1.

Ex 5: Describe geometrically what T does to each vector **u** and **v** in R^2: $T(x)=\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$, $u=\begin{bmatrix}5\\ 2\end{bmatrix}$, $v=\begin{bmatrix}-2\\ 4\end{bmatrix}$

This is a reflection transformation. What line is are the vectors reflected about?

Ex 6: Describe geometrically what T does to each vector **u** and **v** in R^2: $T(x)=\begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$, $u=\begin{bmatrix}5\\ 2\end{bmatrix}$, $v=\begin{bmatrix}-2\\ 4\end{bmatrix}$

This is a shear transformation. It maps line segments onto line segments of different lengths.

Linear Transformations

Definition A transformation (or mapping) T is linear if:

(i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all **u**, **v** in the domain of T

(ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all **u** and all scalars c.

Every matrix transformation is a linear transformation.

A transformation preserves the operations of vector addition and scalar multiplication.

In order to prove that the transformation $T(x_1,x_2) = (x_1 - x_2,\ x_1+2\ x_2\ )$ is a linear transformation, you must first define **u** and **v**.

Let **u** = (u_1, u_2) and **v** = _____

Show the two conditions are true.
1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (start with the left and get to the right)

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T((u_1, u_2) + (v_1, v_2)) \\ &= T(u_1 + v_1, u_2 + v_2) \\ &= ((u_1+v_1)-(u_2+v_2),(u_1+v_1)+2(u_2+ v_2)) \\ &= ((u_1-u_2)+(v_1-v_2),(u_1+2u_2)+(v_1+ 2v_2)) \\ &= (u_1-u_2, u_1+2u_2\ ) + (v_1-v_2,(v_1+ 2v_2) \\ &= T(\mathbf{u}) + T(\mathbf{v}) \end{aligned}$$

2. Show $T(c\mathbf{u}) = cT(\mathbf{u})$ *Complete the proof.*

Ex 7: Prove that the transformation $T(x_1,x_2) = (x_1+ 2\ x_2,3\ x_1- x_2\ )$ is a linear transformation from R^2 to R^2

Ex 8: Prove that the transformation $T(x_1,x_2) = (x_1+1,\ x_2)$ is not a linear transformation from R^2 to R^2

Ex 9: Prove whether the transformation $T(x_1) = 2x_1 + 1$ is a linear transformation.

1.9 The Matrix of a Linear Transformation

We want to be able to write "formula" for any geometric transformation $T(\mathbf{x})$.

Every linear transformation from R^n to R^m is actually a matrix transformation $\mathbf{x} \rightarrow A\mathbf{x}$

The key to finding A is to observe that T is determined by what it does to the columns of the n x n identity matrix I_n.

Identity matrix I_n is an n x n matrix with 1's on the main left to right diagonal and 0's elsewhere. The ith column of I_n is labeled $\mathbf{e}_i$

For example, $I_3 = [\mathbf{e}_1\ \mathbf{e}_2\ \mathbf{e}_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

How do we find T using I_n?

Consider the following. Suppose you know that T is a linear transformation from R^2 to R^3 where $T(\mathbf{e}_1) = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$ and $T(\mathbf{e}_2) = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$ and

that you are asked to find $T(\mathbf{x})$ for any $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

You know that any vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ in R^2 can be written as $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 = x_1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Then $T(\mathbf{x}) = T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2)$

So, $x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) = x_1\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} + x_2\begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2x_1 + 5x_2 \\ _____ \\ _____ \end{bmatrix}$ Thus, the transformation matrix is $\begin{bmatrix} 2 & 5 \\ __ & __ \\ __ & __ \end{bmatrix}$

Note: To get A, replace the identity matrix with the $[\mathbf{e}_1\ \mathbf{e}_2]$ with $[T(\mathbf{e}_1)\ \ T(\mathbf{e}_2)]$

Theorem 10 – Let $T: R^n \rightarrow R^m$ be a linear transformation. Then there exists a unique matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$ in R^n.

In fact, A is the m x n matrix whose j^{th} column is the vector $T(\mathbf{e}_j)$ where $\mathbf{e}_j$ is the jth column of the identity matrix in R^n; $A = [T(\mathbf{e}_1) \ldots T(\mathbf{e}_n)]$

Matrix A is called the standard matrix for the linear transformation T.

Linear transformation focuses on a property of mapping.

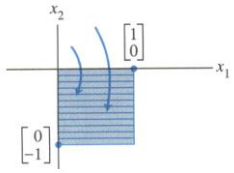
Matrix transformation describes how such a mapping is implemented.

Ex 1: Find the standard matrix of $T: R^2 \rightarrow R^2$ that is a horizontal shear transformation that leaves $\mathbf{e}_1$ unchanged and maps $\mathbf{e}_2$ into $\mathbf{e}_2 + 3\mathbf{e}_1$.

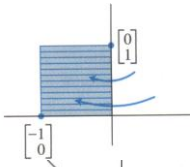
Ex 2: Find the standard matrix of $T: R^2 \rightarrow R^2$ that first reflects points through the horizontal axis x_1, and then reflects points through the line $x_2 = x_1$.

Standard Matrices: Reflections

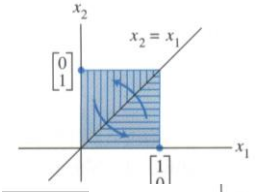
1. Through x_1 axis $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$



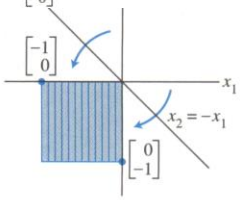
2. Through x_2 axis $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$



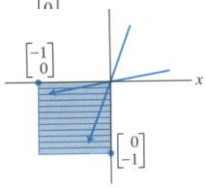
3. Through $x_2 = x_1$ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$



4. Through $x_2 = -x_1$ $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

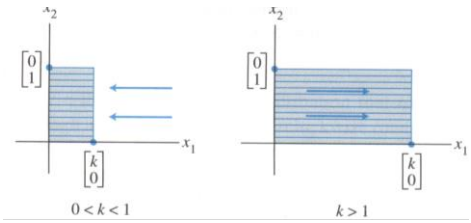


5. Through origin $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

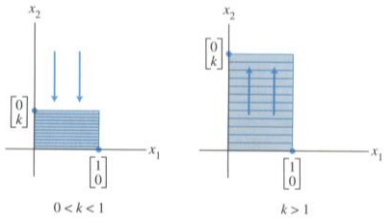


Standard Matrices: Contractions and Expansions

1.Horizontal $\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$

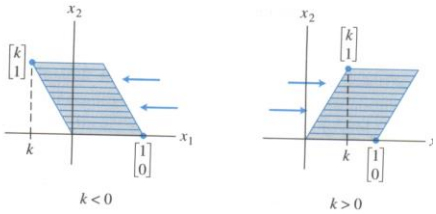


2. Vertical $\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$

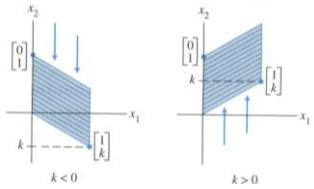


Standard Matrices: Shears

1. Horizontal $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$

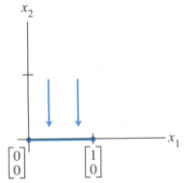


2. Vertical $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

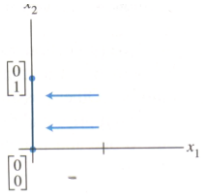


Standard Matrices: Projections

1. x_1 axis $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$



2. x_2 axis $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$



Definition: A mapping $T: R^n \rightarrow R^m$ is said to be onto R^m if each **b** in R^m is the image of at least one **x** in R^n .

- T is onto R^m when the range of T is all of the codomain R^m.
- T maps R^n onto R^m if, for each **b** in the codomain R^m, there exists at least one solution of $T(\mathbf{x}) = b$.
- Mapping T is not onto when there is some **b** in R^m for which the equation $T(\mathbf{x}) = \mathbf{b}$ has no solution.

Definition: A mapping $T: R^n \rightarrow R^m$ is said to be one-to-one if each b in R^m is the image of at most one **x** in R^n

- T is one-to-one if, for each **b** in R^m, the equation $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or none at all.
- The mapping T is not one-to-one when some **b** in R^m is the image of more than one vector in R^n. If there is no such **b**, then T is one-to-one.

Ex 3: Show that T is a linear transformation by finding a matrix that implements the mapping. (x_1,x_2, x_3 are not vectors but entries).
$T(x_1, x_2, x_3) = (x_1 - 5x_2 + 4x_3, x_2 - 6x_3)$

Ex 4: Determine if $T(x_1, x_2, x_3) = (x_1 - 5x_2 + 4x_3, x_2 - 6x_3)$ is (a) one-to-one and (b) onto. Justify each answer.

Ex 5: Let T be the linear transformation whose standard matrix is $\begin{bmatrix} -5 & 10 & -5 & 4 \\ 8 & 3 & -4 & 7 \\ 4 & -9 & 5 & -3 \\ -3 & -2 & 5 & 4 \end{bmatrix}$.

Is T a one-to-one mapping? Justify your answer.

Theorem 11: Let $T: R^n \rightarrow R^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Theorem 12: Let $T: R^n \rightarrow R^m$ be a linear transformation and let A be the standard matrix for T. Then:

- T maps R^n onto R^m if and only if the columns of A span R^m.
- T is one-to-one if and only if the columns of A are linearly independent.

Ex 6: $\begin{bmatrix} 1 & 4 & -5 \\ 3 & -7 & 4 \end{bmatrix}$ Is the standard matrix (a) one-to-one and (b) onto. Justify your answers.

Ex 7: Given $A = \begin{bmatrix} 9 & 13 & 5 & 6 & -1 \\ 14 & 15 & -7 & -6 & 4 \\ -8 & -9 & 12 & -5 & -9 \\ -5 & -6 & -8 & 9 & 8 \\ 13 & 14 & 15 & 2 & 11 \end{bmatrix}$

T is the linear transformation whose standard matrix is given. Does T map R^5 onto R^5? Justify your answer.

1.10 Linear Models in Business, Science, and Engineering

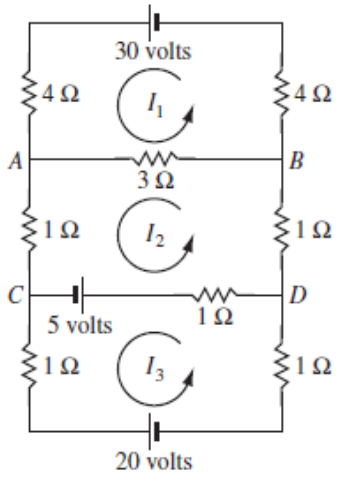
The mathematical models in this section are all *linear*; that is, each describes a problem by means of a linear equation, usually in vector or matrix form. The first model concerns nutrition but actually is representative of a general technique in linear programming problems. The second model comes from electrical engineering. The third model introduces the concept of a *linear difference equation*, a powerful mathematical tool for studying dynamic processes in a wide variety of fields such as engineering, ecology, economics, telecommunication, and the management sciences. Linear models are important because natural phenomena are often linear or nearly linear when the variables involved are held within reasonable bounds. Also, linear models are more easily adapted for computer calculation than are complex nonlinear models.

Ex 1: The formula for the Cambridge Diet, a popular diet in the 1980s, was based on years of research. The very low-calorie powdered formula diet combines a precise balance of carbohydrate, high-quality protein, and fat, together with vitamins, minerals, trace elements, and electrolytes. Millions of persons have used the diet to achieve rapid and substantial weight loss. The following example illustrates the problem on a small scale. Listed in the table are three of the ingredients in the diet, together with the amounts of certain nutrients supplied by 100 grams of each ingredient. If possible, find some combination of nonfat milk, soy flour, and whey, to provide the exact amounts of protein, carbohydrate, and fat supplied by the diet in one day.

Amounts (g) Supplied per 100 g of Ingredient				Amounts (g) Supplied by
Nutrient	**Nonfat milk**	**Soy flour**	**Whey**	**Cambridge Diet in One Day**
Protein	36	51	13	33
Carbohydrate	52	34	74	45
Fat	0	7	1.1	3

Ex 2: Current flow in a simple electrical network can be described by a system of linear equations. A voltage source such as a battery forces a current of electrons to flow through the network. When the current passes through a resistor (such as a lightbulb or motor), some of the voltage is "used up"; by Ohm's law, this "voltage drop" across a resistor is given by $V = IR$, where voltage V is measured in volts, the resistance R in ohms (denoted by the Greek letter Ω), and the current flow I in amperes (amps, for short). The network in the figure contains three closed loops. The currents flowing in loops 1, 2, and 3 are denoted by I_1, I_2, and I_3, respectively. The designated directions of such *loop currents* are arbitrary. If a current turns out to be negative, then the actual direction of current flow is opposite to that chosen in the figure. If the current direction shown is away from the positive (longer) side of a battery (—|—) around to the negative (shorter) side, the voltage is positive; otherwise, the voltage is negative. Current flow in a loop is governed by Kirchhoff's Voltage Law, which says that the algebraic sum of the IR voltage drops in one direction around a loop equals the algebraic sum of the voltage sources in the same direction around the loop.

Use this information to determine the loop currents in the network shown.



Ex 3: A subject of interest to demographers is the movement of populations or groups of people from one region to another. The simple model here considers the changes in the population of a certain city and its surrounding suburbs over a period of years. Fix an initial year – say, 2000 – and denote the populations of the city and suburbs that year by r_0 and s_0, respectively. Let $\mathbf{x_0}$ be the population vector, and for 2001 and subsequent years, denote the populations of the city and suburbs as shown:

$$x_0 = \begin{bmatrix} r_0 \\ s_0 \end{bmatrix}, \qquad x_1 = \begin{bmatrix} r_1 \\ s_1 \end{bmatrix}, \qquad x_2 = \begin{bmatrix} r_2 \\ s_2 \end{bmatrix}, \text{ etc.}$$

Our goal is to describe mathematically how these vectors might be related. Suppose demographic studies show that each year about 5% of the city's population moves to the suburbs (and 95% remains in the city), while 3% of the suburban population moves to the city (and 97% remains in the suburbs). After 1 year, the original r_0 persons in the city are now distributed differently between the city and suburbs, as are the original s_0 persons in the suburb, giving the equation:

$$\begin{bmatrix} r_1 \\ s_1 \end{bmatrix} = r_0 \begin{bmatrix} .95 \\ .05 \end{bmatrix} + s_0 \begin{bmatrix} .03 \\ .97 \end{bmatrix} = \begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix} \begin{bmatrix} r_0 \\ s_0 \end{bmatrix}$$

The 2x2 matrix is defined as the **migration matrix**, and the equation $x_2 = Mx_1$ describes how the population changes from 2000 to 2001. If the migration percentages remain constant, the sequence of x_k vectors describes the population of the city/suburban region over a period of years, where $x_{k+1} = Mx_k$.

Compute the population of the region just described for the years 2001 and 2002, given that the population in 2000 was 600,000 in the city and 400,000 in the suburbs.